Mandatory Assignment 1

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Exercise 1

We download monthly data for the S&P 500 index (`^GSPC`) and the 13-week Treasury Bill rate (`^IRX`) from Yahoo! Finance for the period January 1, 2000 to February 28, 2026. The `^IRX` series is an annualized yield in percent, so it is converted into a monthly risk-free rate as

$$r_{f,t} = (1 + IRX_t/100)^{1/12} - 1.$$

Monthly excess market returns are then computed as

$$\tilde{r}_{m,t} = r_{m,t} - r_{f,t},$$

where $r_{m,t}$ denotes the monthly return on the S&P 500 index. We use the estimated monthly mean excess return μ_m and monthly variance σ_m^2 to calibrate the synthetic asset universe used in the remainder of the assignment.

Table 1: Estimated monthly and annualized market parameters.

Parameter	Monthly	Annualised
Mean excess return	0.0045	0.0545
Variance	0.0019	0.0228

Note: Based on monthly S&P 500 excess returns (`^GSPC`) from January 2000 to February 2026. The risk-free rate is `^IRX`, converted to a monthly rate.

Table 1 reports the estimated monthly mean excess market return and variance, together with their annualized counterparts. The estimated monthly mean excess return is approximately 0.45%, while the annualized mean excess return is approximately 5.45%. The corresponding annualized variance is approximately 0.0228.

Exercise 2

Next, we construct a universe of $N = 50$ synthetic assets following the single-factor model

$$\tilde{r}_{i,t} = \beta_i \tilde{r}_{m,t} + \varepsilon_{i,t}.$$

The factor loadings β_i and idiosyncratic volatilities $\sigma_{\varepsilon,i}$ are drawn from the distributions specified in the assignment. Using the estimated market moments from Exercise 1, we construct the true mean vector μ and covariance matrix Σ, which are treated as the true parameters throughout the remainder of the assignment.

Table 2: Summary statistics for the synthetic asset universe.

Parameter	Min	Mean	Max
Beta	0.523	0.968	1.4876
Idiosyncratic volatility	0.029	0.0604	0.0865
Expected excess return	0.0024	0.0044	0.0068

Note: Summary statistics for the N = 50 synthetic assets.
All quantities are reported in monthly units.

Table 2 summarizes the characteristics of the $N = 50$ synthetic assets. The true covariance matrix is constructed as

$$\Sigma = \sigma_m^2 \beta\beta' + \mathrm{diag}(\sigma_\varepsilon^2),$$

where the first term captures systematic market risk and the second term captures idiosyncratic risk.

Exercise 3

Using the true mean vector μ and covariance matrix Σ constructed in Exercise 2, we implement functions to compute the tangency portfolio and the mean–variance efficient frontier.

The efficient frontier consists of portfolios that minimize portfolio variance for a given target return,

$$\omega_{eff}(\bar{\mu}) = \arg\min_{\omega} \omega' \Sigma \omega,$$

subject to

$$\omega' \iota = 1, \qquad \omega' \mu \geq \bar{\mu}.$$

The tangency portfolio is computed as

$$\omega_{tan} = \frac{\Sigma^{-1}\tilde{\mu}}{\iota' \Sigma^{-1} \tilde{\mu}},$$

where $\tilde{\mu} = \mu - r_f$ denotes expected excess returns.

The implemented functions return portfolio weights together with the corresponding expected return, variance, volatility and Sharpe ratio. They are used throughout the remainder of the assignment to evaluate portfolio allocations under both true and estimated parameters.

Exercise 4

Using the true parameter values μ and Σ, we compute the mean-variance efficient frontier and the corresponding tangency portfolio. The resulting frontier describes the best attainable risk-return combinations under the true data-generating process.

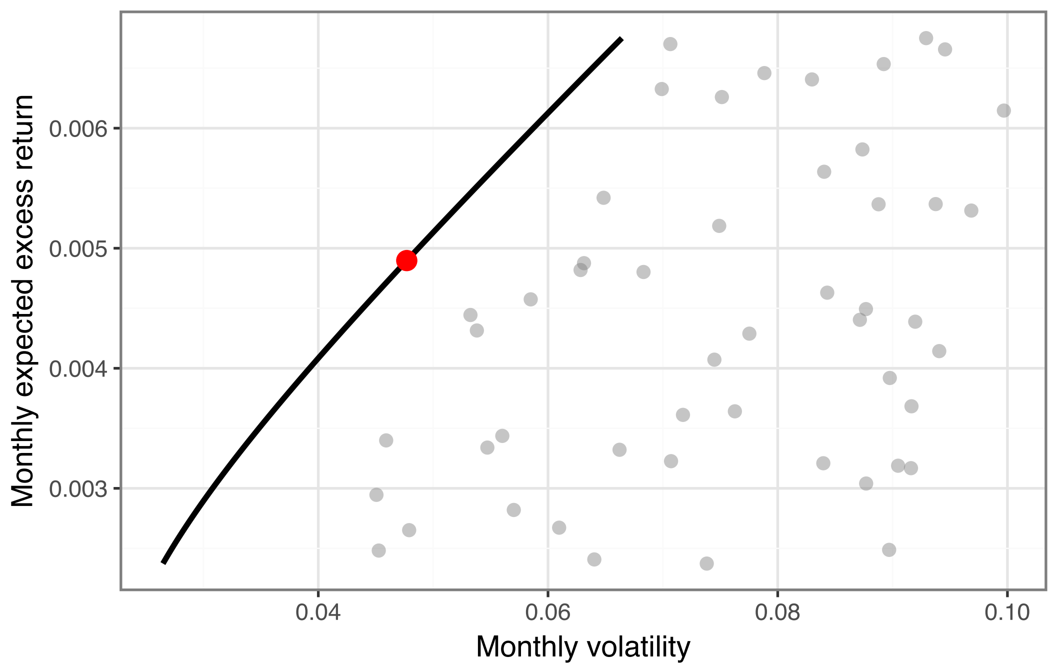

Figure 1: Mean-variance efficient frontier and tangency portfolio based on the true parameters.

Figure 1 shows the risk-return characteristics of the 50 synthetic assets together with the mean-variance efficient frontier. The grey points represent individual assets, while the frontier shows the set of portfolios that achieve the highest expected excess return for a given level of risk. The marked point denotes the tangency portfolio, which maximizes the Sharpe ratio and therefore represents the most attractive risk-return trade-off under the true parameter values.

Exercise 5

To study estimation error, we simulate asset returns from the true data-generating process and estimate the mean vector $\hat{\mu}$ and covariance matrix $\hat{\Sigma}$ from the simulated sample.

Using this function, we repeatedly simulate returns (30 times) and construct plug-in efficient frontiers based on the estimated parameters $\hat{\mu}$ and $\hat{\Sigma}$. We consider sample sizes of $T = 60$, $T = 120$

and $T = 500$ to examine how the magnitude of estimation error depends on the amount of available data.

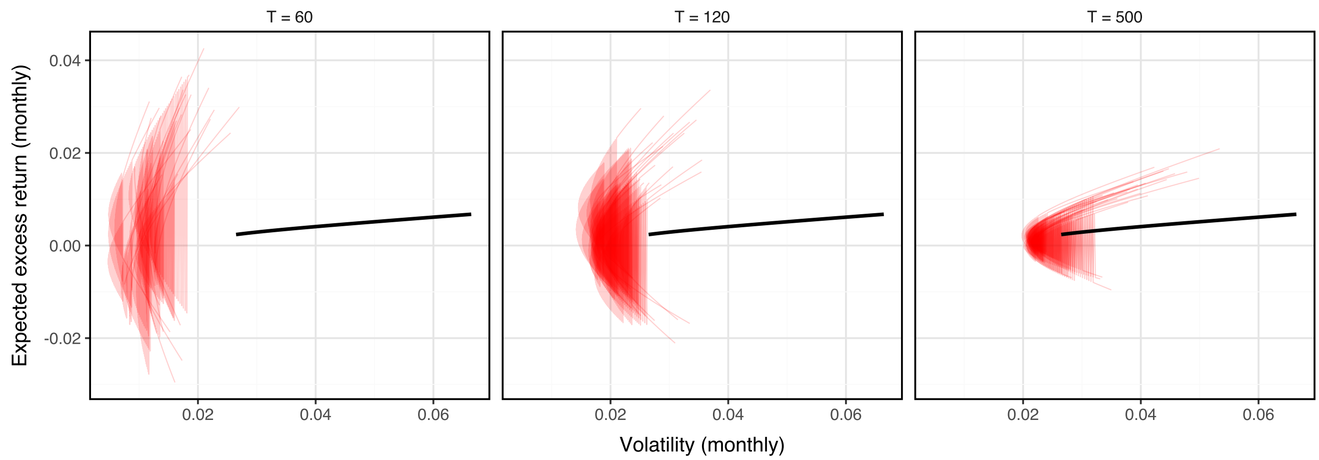

Figure 2: Estimated efficient frontiers from repeated simulations compared with the true frontier.

Figure 2 compares the true efficient frontier (black) with frontiers estimated from simulated samples (red) of different lengths. For T=60, the estimated frontiers display substantial variation across simulations and often differ markedly from the true frontier. As the sample size increases to T=120 and T=500, the estimated frontiers become increasingly clustered around the true frontier. This indicates that larger samples produce more reliable estimates of the mean vector and covariance matrix, leading to more accurate estimates of the efficient frontier. Overall, the figure shows that estimation error becomes less severe as the amount of available data increases.

Exercise 6

To evaluate the impact of estimation error on portfolio performance, we compare the true tangency portfolio with the plug-in tangency portfolio. The plug-in portfolio is constructed from the estimated moments $\hat{\mu}$ and $\hat{\Sigma}$, but its Sharpe ratio is evaluated using the true parameters μ and Σ.

The out-of-sample Sharpe ratio is computed as

$$SR(w) = \frac{w^\top \mu}{\sqrt{w^\top \Sigma w}}.$$

Table 3: Out-of-sample Sharpe ratios of the plug-in tangency portfolio.

T	Mean plug-in Sharpe	Std plug-in Sharpe	True Sharpe	Sharpe loss
60	0.0062	0.0139	0.1027	0.0965
120	0.0081	0.0169	0.1027	0.0946
240	0.015	0.0189	0.1027	0.0876
500	0.023	0.0224	0.1027	0.0797

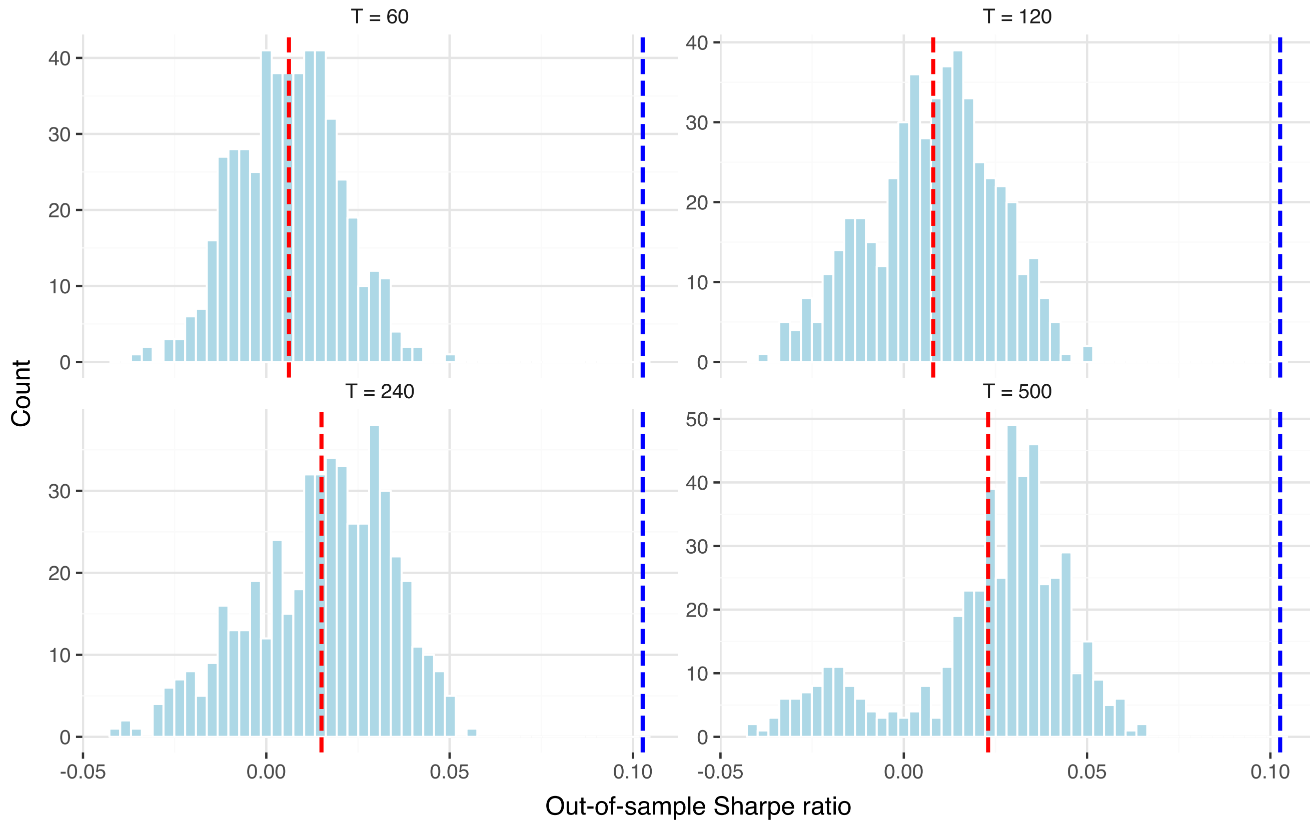

Figure 3: Distribution of out-of-sample Sharpe ratios for the plug-in tangency portfolio.

Table 3 and Figure 3 show the out-of-sample performance of the plug-in tangency portfolio. The blue dashed line marks the Sharpe ratio of the true tangency portfolio, while the red dashed line marks the average plug-in Sharpe ratio for each sample size. The histograms show the distribution of plug-in Sharpe ratios across simulations. When (T) is small, the plug-in Sharpe ratios are dispersed and lie far below the true optimum. As (T) increases, the distribution shifts closer to the true Sharpe ratio, showing that performance improves when the sample moments are estimated more precisely.

Exercise 7

In Exercise 6, we focused on the plug-in tangency portfolio and examined how estimation error affects its out-of-sample performance. We now extend the analysis by comparing the plug-in tangency portfolio with alternative portfolio construction methods. Using the simulation framework from Exercise 5, returns are generated from the true parameters μ and Σ. Portfolio weights are then constructed from the simulated samples and evaluated out-of-sample using the true parameters. This allows us to assess which strategies are most robust to estimation error.

The benchmark strategy is the equally weighted portfolio, defined as

$$w_{EW} = \frac{1}{N}\iota.$$

This portfolio assigns equal weights to all assets and does not depend on estimated parameters. The economic intuition is that diversification can be achieved without attempting to identify the best assets.

The first alternative strategy is the global minimum variance portfolio, which minimizes portfolio variance subject to the weights summing to one. The weights are given by

$$w_{GMV} = \frac{\Sigma^{-1}\iota}{\iota^{\top}\Sigma^{-1}\iota}.$$

This strategy depends only on the covariance matrix and does not require estimates of expected returns.

The second alternative strategy is the inverse-volatility portfolio, where assets with higher volatility receive lower weights. The weights are defined as

$$w_i = \frac{1/\sigma_i}{\sum_{j=1}^{N} 1/\sigma_j}.$$

This strategy is simple and only relies on the individual asset volatilities.

The plug-in tangency portfolio from Exercise 6 is included as a reference strategy. It uses both $\hat{\mu}$ and $\hat{\Sigma}$, and is therefore expected to be more sensitive to estimation error.

Table 4: Average out-of-sample Sharpe ratios across portfolio strategies and sample sizes.

strategy T	Equally weighted	Global minimum variance	Inverse volatility	Plug-in tangency
60	0.1018	0.0206	0.1020	0.0058
120	0.1018	0.0364	0.1020	0.0081
240	0.1018	0.0425	0.1020	0.0162
500	0.1018	0.0454	0.1020	0.0235
True optimum	0.1027	0.1027	0.1027	0.1027

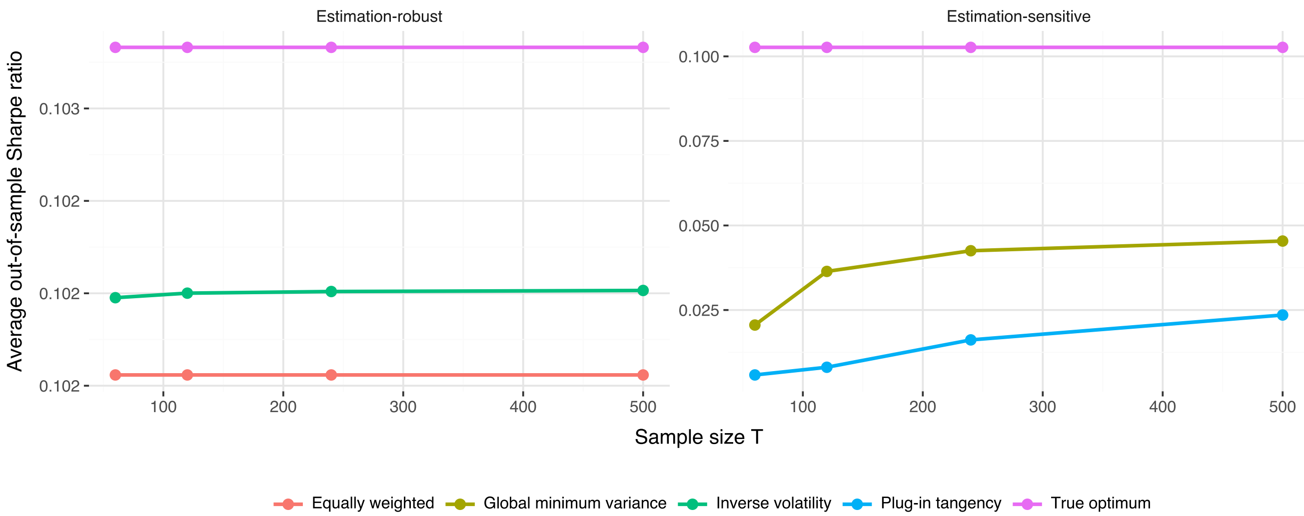

Figure 4: Out-of-sample Sharpe ratios across portfolio strategies.

Table 4 and Figure 4 compare the average out-of-sample Sharpe ratios across portfolio strategies and sample sizes. The true optimum is included as a benchmark representing the highest Sharpe ratio attainable when the true parameters are known.

The left part of the figure displays the estimation-robust strategies, namely the equally weighted portfolio and the inverse-volatility portfolio. Both strategies achieve Sharpe ratios that are very close to the true optimum and remain almost unchanged as the sample size increases. This indicates that their performance is largely unaffected by estimation error because they rely little or not at all on estimated parameters.

The right part of the figure displays the estimation-sensitive strategies, namely the plug-in tangency portfolio and the global minimum variance portfolio. Both strategies perform substantially worse than the true optimum when the sample size is small. As the sample size increases, their Sharpe ratios improve because the estimates of the covariance matrix and expected returns become more precise. The improvement is particularly pronounced for the plug-in tangency portfolio, which depends on estimates of both $\hat{\mu}$ and $\hat{\Sigma}$.

Overall, the results suggest that the equally weighted and inverse-volatility portfolios are the most robust to estimation error, while the plug-in tangency portfolio is the most sensitive. Although optimization-based strategies are theoretically attractive, their performance can deteriorate considerably when the inputs are estimated with noise.

A practical lesson for investors is therefore that simpler portfolio rules may outperform highly optimized portfolios in realistic settings where parameters must be estimated from limited historical data. The analysis highlights that reducing estimation risk can be just as important as achieving theoretical optimality, especially when sample sizes are small.